FACULTY OF NATURAL SCIENCE
DEPARTMENT OF COMPUTER SCIENCE

CSE 2101: Software Engineering

Software Requirements Specification: Student Records Management System (SRMS) Mobile Application

Client: Software Services & Educational Technology Applications (SSETA)

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Group 5

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1.0 Introduction

1.1 Purpose

This Software Requirements Specification document describes the requirements for the Student Records Management Systems (SRMS) Mobile application, a mobile extension of the current University of Guyana's SRMS web application.

The SRMS Mobile Application is intended to provide students with timely updates and quick access to their university information directly from their mobile devices. The major goal of the system is to provide a modern, fast, and convenient experience for students which will enable them to manage an essential part of their academic life on the move all from their pocket.

The app will support major functions which are currently available on the SRMS website including student application, timetable viewing, results access, and system notification, while making sure the system has a cleaner and more efficient user interface. The SRS focuses on student related features for the initial launch, with the prospect of expanding to include limited staff interaction for later updates.

Every action and updates performed through the mobile application should be fully synchronized with the existing SRMS backend. This guarantees that updates made on the web or the application are immediately mirrored across both platforms. This SRS document outlines the system's intended behavior, its functional and non-functional requirements and constraints affecting the design.

1.2 Document Conventions

The following conventions are used throughout this document to ensure clarity and consistency:

- **FR-X -** Functional Requirement (where the prefix denotes the feature, e.g. CR-X for **C**ourse **R**egistration)
- **NFR-X -** Non-Functional Requirement
- **UC-XX -** Use Case

Priority Levels:

- **High -** Essential for core system operation
- **Medium -** Important features that improve the system but are not essential for system operation.
- **Low -** Optional features that may be implemented in future releases.

Additionally, all models and diagrams used in this document follow standard software engineering notation, specifically the Unified Modeling Language (UML) for Use Case diagrams.

1.3 Intended Audience

This SRS document is intended for several different groups of people involved in understanding, analyzing or supporting the development of the SRMS Mobile Application. Each stakeholder may have a different level of understanding and a different reason for using the document:

- **Client/ Project Sponsor (Mr. Tyreke Wilson) -** Evaluates the requirements and ensures that the systems match the intended goal of the University of Guyana.
- **Development Team-** Uses the document to guide requirement analysis, system modeling, presentation and future design decisions.
- **Tester/ Quality Assurance Personnel -** Uses this SRS document to develop test cases and to verify that the final application meets functional and non-functional requirements.
- **UG ICT department -** Uses or refer to this document for backend integration, data handling, and long term maintenance of the system.
- **Future Developers and Maintainers -** Uses the document for reference when expanding or deploying the mobile application.

1.4 Reading Suggestions

To make traversing easier, the document is organized into various section that goes from a high level system understanding to detailed requirements:

Section 1: Introduces the purpose of the system and identifies the target audience.

Section 2: Provides a general overview of the system, including product perspective, user characteristics and constraints.

Section 3: Outlines the detailed functional requirements, system features and use cases. This is the most important section for developers and testers.

Section 4: Details the non-functional requirements, including performance, security and quality attributes.

2.0 Overall Description

2.1 Product Perspective

The SRMS mobile application is a new addition to the University of Guyana's current student records management system. Students can only access SRMS through the website which was designed mainly for access on computers and not phones. Because of this, many students struggle to use the system on their mobile devices. The screen does not adjust well, pages take long to load and the website often overloads because of traffic during busy times like course registration.

The mobile app is being developed to solve all these problems. Students will be able to access their academic information from their smartphones in a cleaner and more efficient way. This will not replace the web version. It will work together with it and use a new API layer that connects directly to the same SRMS database so the information stays updated on both platforms.

The first release of the mobile application will focus mainly on student features while future updates may include limited tools for staff members. According to the client, the SRMS web application is built on an older PHP based structure so to ensure correct synchronization between both applications, the mobile app must communicate with the web application through a new middleware system that the developers will create.

2.2 Product Functions

The SRMS mobile application will allow current and prospective students to access important academic features from their mobile devices. The main functions of the system will include:

1. **Student admission and application-** This will allow users to create a new applicant account, submit applications, upload supporting documents (ID, certificates, etc) and track the progress and status of their application.
2. **Course registration-** Current students will be able to view available courses for the semester, register for and drop courses, view their registered course list in real time and receive confirmation messages for registration actions.
3. **Timetable viewing-** This function will display students' weekly class timetable, automatically update the timetable after registration changes and show class times, class method, room numbers and course codes.
4. **Academic results-** At the end of the semester, students will be able to view their exam grades and GPA and also receive notifications whenever new results are released.

5. **Notifications and alerts-** The mobile app should have a feature where students receive push notifications directly to their smartphones when the registration period is open, new grades are released, there is an update on their applications and any system announcements.
6. **Payment processing for foreign students-** All students who are not Guyanese citizens are required to pay for their tuition. They should be able to view their balance, view generated invoices and make online payments securely.
7. **User account and profile management-** All students should be able to login securely using their SRMS credentials, reset their password, update contact information and use two factor authentication for sensitive actions.
8. **Real-time synchronization-** The system should be able to update the SRMS web system instantly when any actions are done on the mobile app and vice versa. This ensures data consistency across both platforms.

2.3 User Classes and Characteristics

The SRMS mobile application will be used by several groups of people with different needs, level of experience and system access. To build an effective and usable application, developers will have to understand each of these user classes and their requirements.

1. **Local Guyanese Students-** Majority of the users will be local Guyanese citizens who will be using the app frequently during registration, exam periods and for timetable checks. A lot of persons live in areas with fluctuating internet connectivity (3G/4G) and use older mobile devices with limited storage. Also, persons have different technical knowledge. Some are highly skilled and others only know basic smartphone use.
2. **Foreign Students-** Foreign students must be able to access all general features as local students. Additionally, they must be able to access the payment system since only Guyanese citizens get free tuition. The mobile app must have a secure and reliable financial processing system in place. Also foreign students may depend heavily on the app for updates while they are overseas.
3. **Prospective Students-** These are people not yet enrolled in the university. They usually have limited knowledge of UG systems. They would be using the app mainly during application periods and require access to only a small part of the system. Their needs are simpler than students who are enrolled but it is still important for recruitment.

4. **UG Registry Department Staff-** These are more experienced persons using the system. They require higher privilege access and work with sensitive and personal data. They usually use desktop systems more than mobile devices. Phase 1 focuses on student-facing features but staff administrative features are scheduled for Phase 2 so the system must prepare for staff access in later updates.
5. **UG ICT Department/System Administrators-** These persons have advanced technical knowledge. They are responsible for maintenance, troubleshooting and server/API performance. They need access to logs, API monitoring tools and data synchronization reports. They are the internal support users so their interface and features will be handled separately.

2.4 Design and Implementation Constraints

The development of the SRMS mobile application will face several constraints that limit how the system can be designed and implemented.

1. **Existing SRMS architecture-** The current SRMS is built on an older PHP based system with no API layer. The mobile app must connect to the SRMS through a new API which the team must build from scratch. Because of this the design must match the old database structure even if it is not modern or flexible. This will limit how fast new features can be added and will need extra testing to prevent data mismatches.
2. **Hardware limitations-** Some UG students use older or lower spec mobile devices (2gb ram or less). The app shall not exceed 50MB in order to load on older phones. It must work on Android 8 and above and iOS 14 and above. This limits heavy graphics, large libraries or very complex animations.
3. **Performance constraints-** The app must be able to handle high traffic especially during course registration periods. Therefore, the servers should support up to 20000+ students using the system at the same time and loading should take under 3 seconds even on slow 3G/4G networks.
4. **Security and legal requirements-** The app must follow Guyana Data Protection Act and basic privacy rules. It also needs to meet the GDPR standards since UG had some foreign students. This requires the system to have strong data encryption, secure user authentication and proper handling of documents. Because of this, the system must use TLS 1.3 for secure data transfer, AES-256 for encrypted storage and two factor authentication for sensitive actions. This will prevent weak security libraries and outdated encryption methods.

5. **Payment system requirements-** Only foreign students need to pay tuition so the app should have the correct features to identify who needs that feature. The system must connect to an external online payment gateway which has its own rules and restrictions. The payment gateway may require secure merchant APIs, currency and invoice formatting and verified student identity. This limits how payment features can be designed and when they can be activated.
6. **Communication and integration constraints-** The mobile app must stay synchronized with the web based SRMS at all times. Both systems will be using the same database so any update must appear instantly. If the web SRMS goes down, the mobile app will also be affected. This limits offline features and requires strong error handling.
7. **Waterfall development process constraint-** The project uses the plan driven waterfall model which means requirements cannot be changed later, each phase must be finished before moving on and no major rework can happen after the development of the project starts. Flexibility is limited and accurate planning is required.
8. **Platform and store requirements-** The app must meet google play store and apple store rules such as app size, privacy policy, security standards and screenshot and icon specifications. This limits design choices and requires extra documentation.

2.5 User Documentation

The SRMS mobile application will include several types of user documentation to help students, applicants and support staff understand how to use the system. It must be easy to understand and follow to cater for first time users.

1. **In-app help guide-** the mobile app will include a built in help section that explains how to use the main features. This guide will include things like step by step instructions for registration, how to view grades how to track application status, how to find your timetable and what to do if you get an error. This can be accessed from the app's menu.
2. **Quick start guide PDF-** this is a short beginner friendly pdf which will be provided to new users and will include how to log in, how to reset password, overview of the home screen, icons and what they mean and basic troubleshooting tips.
3. **Full user manual-** a more detailed user manual will be created to explain every function of the app. It will include screenshots of each page, how each feature works, common issues and solutions and how to contact the UG ICT helpdesk. This is for students who run into serious problems or want more details.

4. **Video tutorials-** short video guides can be created to show users how to apply to UG, register for courses, view grades and use the payment section for foreign students. This may help those who are visual learners.
5. **App store descriptions-** once the app is submitted to google play and apple app store, each listing will include a brief summary of what the app does, screenshots, privacy information and contact details for support. This helps users understand the app before downloading it.
6. **Support and helpdesk information-** The app will include contact information for UG ICT helpdesk and the registry department for application and registration issues. Students can get help through email, phone or on campus support during work hours.

2.6 Assumptions and Dependencies

The development of the SRMS mobile application is based on several assumptions about technology, users and the university's existing system. The project also depends on external components and services that must work correctly for the app to function.

2.6.1 Assumptions

1. **Reliable access to the existing SRMS system-** it is assumed that the University of Guyana will give the development team full access to the current SRMS database and backend so the app can connect through the new API but if this cannot be done, the team may not be able to fully integrate the mobile app.
2. **Students have internet access-** we assume that students and applicants have mobile data or Wi-Fi to use the app. If internet access is poor, some features like real time syncing may not work properly.
3. **UG ICT department will maintain servers-** we assume that the ICT department will host the API, maintain server security and scale resources during busy periods like registration. If server support is delayed, performance issues could occur.
4. **Students will prefer the mobile app over the website-** we assume that the mobile version will achieve a user adoption rate of at least 60% of the student population because it offers a better user experience. If there are less complaints about the website, the app's impact may be reduced.
5. **The waterfall model will be followed-** we assume that requirements will remain stable throughout the project since the waterfall method does not allow large changes later. If new requirements appear late, development delays may occur.

2.6.2 Dependencies

1. **API middleware layer-** the entire mobile app depends on the new API layer that will connect it to the existing SRMS. If the API is delayed or fails, main features like registration and results will not work.
2. **Existing SRMS database-** the mobile app depends on the current SRMS database structure. Any changes to the database by the university may affect syncing and performance.
3. **External payment gateway-** payment features depend on an external payment system for foreign students. If the provider changes fees, policies or API settings, the mobile app must be updated.
4. **Push notification service-** the mobile app depends on an external notification service to send alerts such as grade updates, registration reminders, and system announcements. If this notification service has problems or downtime, alerts may be delayed or may not reach users on time.
5. **Mobile app stores-** deployment depends on meeting the requirements of google play store and apple app store. If either store rejects the app, launch dates may be affected.
6. **UG ICT Infrastructure-** The app relies on the university's server uptime, cybersecurity measures, database access and network stability. Any ICT outage directly affects the app.

3.0 System Features

3.1 Course Registration

3.1.1 Description and Priority

The Course Registration feature enables students to register for courses, make changes to their registration and view the list of courses they have registered for the current semester. The mobile application should retrieve all necessary data from the SRMS database, through the unified API and be fully synchronized with the web application.

Priority: High

This feature is critical to the system operations. Students will depend on the registration process to get access to their classes, view timetables and receive their academic credit.

3.1.2 Stimulus/Response Sequences

- **Stimulus:** A student logs into the mobile application and opens the Registration screen.
- **Response:** The system retrieves the student's programme, study year, residency status, their previously registered courses if available, and any outstanding payment requirements.
- **Response:** The system fetches all available courses for the student's programme and semester including required, elective, and additional courses.
- **Stimulus:** The student chooses a course to add or drop.
- **Response:** The system sends a request to the SRMS API.
- **Response:** The API verifies the prerequisites, residency/payment rules, and programme requirements.
- **Response:** If the request is valid, the course selection is updated and is reflected in the provisional registration list.
- **Stimulus:** The student moves to a preview screen and selects to submit their registration.
- **Response:** The system sends the final submission to the SRMS API.
- **Response:** The system sends a notification upon the update of the registration status and displays the registration status (e.g., Online Registration: Submitted, HOD: Approved; Asst. Dean: Approved; Fees due before ARA approval is given).

3.1.3 Functional Requirements

- **CR-1:** The system shall allow all students to view all available courses for their programme and semester, also including course code, course name, credit value, and category (required, elective, or additional).
- **CR-2:** The system shall allow students to select courses through the mobile application and synchronize these selections with the web SRMS via the unified API.

- **CR-3:** The system shall validate each course selection using the necessary courses or SRMS rules, including prerequisites and programme requirements.
- **CR-4:** The system shall prevent students from selecting courses that fail automatic checks and shall display the appropriate error message.
- **CR-5:** The system shall allow students to drop or register for courses within the allowed period only.
- **CR-6:** The system shall enforce different flows based on residency status (e.g., foreign and CARICOM students who are required to make payments before approval).
- **CR-7:** The system shall provide a preview and submit screen showing selected courses, total credits, and key personal information before final submission.
- **CR-8:** The system shall require students to explicitly confirm whether they want to submit their registration before any submission request is sent to the SRMS.
- **CR-9:** The systems shall display a confirmation message once the registration has been successfully submitted.
- **CR-10:** The system shall show the student's official registration status returned by the SRMS.
- **CR-11:** The system shall display the reason for the return or rejected submission when updated by the HOD.
- **CR-12:** The system shall synchronize all registration data between the mobile and the SRMS web application.

3.2 Student Application

3.2.1 Description and Priority

The student Application feature enables prospective students to submit an application to the University of Guyana using the mobile app. This would include creating an applicant account, entering personal details, address details, educational background, employment history (if applicable), choosing a programme of study, uploading supporting documents, and reviewing the completed application before submission.

The mobile application will not replace the current SRMS web application but provide a more user/mobile friendly way to complete the same steps. All application data must be sent and stored in the current SRMS backend via the unified API enabling application to be reviewed as usual with the web application. Any application status updates such as Submitted, Under Review, Return for Changes, Accepted or Rejected must be retrieved from the SRMS to ensure both platforms are synchronized.

Priority: High

This is a core feature for new students trying to enter into the University of Guyana. The submission of an application directly impacts the admission workflow of the registry.

3.2.2 Stimulus/Response Sequences

- **Stimulus:** A prospective student opens the mobile application, selects “Apply to UG”, and starts the online application process.
- **Response:** The system displays the application login screen. If the user doesn’t have an applicant login, the system provides an option to create a new login.
- **Stimulus:** The student logs in and completes the Personal Information form.
- **Response:** The system validates the information before allowing the user to continue.
- **Stimulus:** The student enters mailing and permanent address details.
- **Response:** The system stores the data and confirms that the progress so far has been stored.
- **Stimulus:** The student enters academic qualifications and employment history.
- **Response:** The system stores the information entered.
- **Response:** The system retrieves a list of available programmes from the SRMS and displays them.
- **Stimulus:** The student selects the desired programme and uploads documents.
- **Response:** The uploaded documents are stored and linked to the application.
- **Stimulus:** The student confirms submission.
- **Response:** The system transmits the application data to the SRMS backend using the unified API, displays a confirmation message, and updates the application status.

3.2.3. Functional Requirements

- **SA-1:** The system shall allow a user to login with an existing applicant account or create a new applicant account before starting the process.
- **SA-2:** The system shall allow users to input and save personal information, including full name, date of birth, gender, marital status, country of birth, citizenship and region.
- **SA-3:** The system shall allow users to input and save their mailing and permanent address.
- **SA-4:** The system shall allow users to input and save their academic qualifications, including examination type, subject, grades, and completion dates.
- **SA-5:** The system shall allow users to optionally input and store their employment history information.
- **SA-6:** The system shall fetch the list of all available programmes from SRMS and allow users to select a programme.
- **SA-7:** The system shall allow users to upload the required supporting documents.
- **SA-8:** The system shall validate all required inputs fields and documents before allowing the user to proceed with their submission,

- **SA-9:** The system shall provide a preview and submit screen that displays a summary of the information entered and documents uploaded.
- **SA-10:** The system shall be required to explicitly confirm whether or not they would like to submit, before sending the information to the SRMS backend.
- **SA-11:** The system shall send the complete application and associated documents to the SRMS backend and display a confirmation when the submission is successful.
- **SA-12:** The system shall retrieve and display the current application status as stored in SRMS.
- **SA-13:** The system shall display any comments or reason provided by the Registry staff if the application is returned for changes.

3.3 Timetable Viewing

3.3.1 Description and Priority

The Timetable Viewing feature should enable students to view their class schedule for the current semester. The mobile application retrieves the timetable information from the SMRS through the unified API. The timetable should display the course code, course name, days, times and venue in a clear and legible manner.

This feature would allow students quick access to their weekly schedule which will support better time management and planning.

Priority: Medium

The feature is important for usability and convenience but is not essential for the core registration and submission operation.

3.3.2 Stimulus/Response Sequences

- **Stimulus:** A student logs into the mobile application and selects the Timetable option.
- **Response:** The system fetches the courses which the student is officially enrolled in for the current semester.
- **Response:** The system retrieves the timetable entries for each enrolled course, including day, start time, end time, lecturer(s), and room.
- **Response:** The system displays the timetable in a clean weekly layout or in a suitable format for mobile screens.
- **Stimulus:** The student scrolls or toggles views.
- **Response:** The system updates the display to show different days or switch between weekly/daily views.
- **Response:** If a timetable entry is labeled as “Hybrid”, the system displays a note to the student advising them to make contact with the lecturer for further details.

3.3.3 Functional Requirements

- **TV-1:** The system shall retrieve all courses the student is enrolled in for the current semester .
- **TV-2:** The system shall retrieve the timetable information for each enrolled course, including day, start time, end time, lecturer(s) name, and room.
- **TV-3:** The system shall display the timetable in an appropriate format for mobile devices (weekly or daily).
- **TV-4:** The system shall update the timetable if any official changes are made to the course registration.
- **TV-5:** The system shall support timetable viewing in both weekly and daily formats/views.
- **TV-6:** The system shall display a note explaining that details should be confirmed with the lecturer, if the course is labeled as "Hybrid".
- **TV-7:** The system shall restrict the timetable display to only the student's enrolled courses

3.4 Academic Progress and Records

3.4.1 Description and Priority

The Academic Progress and Records feature is the central dashboard for students to view their academic performance throughout their time at the university. This feature allows students to view their grades for the current semester in detail, access their full academic history and track their cumulative Grade Point Average (GPA).

All academic data must be retrieved in real-time from the central SRMS database via the unified API to ensure that what the student sees on their phone matches exactly what is on the web. Additionally, this feature includes a notification feature to keep students informed when a grade is published.

Constraint: For foreign students, access to this feature is dependent on financial clearance. The system will block access to grades and transcripts until the payment is verified.

Priority: High

This is one of the main reasons a student would install the app. Providing immediate and easy access to grades removes the anxiety of waiting and the need to constantly refresh the web application.

3.4.2 Stimulus/Response Sequences

- **Stimulus:** A student logs into the mobile application and taps the "Academic Record" button on the dashboard.

- **Response:** The system checks the student's financial status (for Foreign/CARICOM students).
- **Response (Scenario A- Cleared):** The system sends a secure request through the API to fetch the student's complete academic history from the SRMS backend.
- **Response:** The system organizes and displays the data by semester, listing the course code, course title and final grade for each entry.
- **Response:** The system calculates and displays the student's current cumulative Grade Point Average (GPA).
- **Response (Scenario B - Not Cleared):** The system displays a notification: "Access to grades is restricted due to an outstanding balance. Please check the Finances tab."
- **Stimulus:** A lecturer or administrator publishes a new grade in the legacy SRMS system.
- **Response:** The system sends a notification to the student's phone that a new grade is posted.

3.4.3 Functional Requirements

- **APR-1:** The system shall retrieve and show a read-only list of all registered courses and final grades for every completed semester.
- **APR-2:** The system shall restrict access to grades and transcripts for Foreign/CARICOM students who have not obtained financial clearance.
- **APR-3:** The system shall calculate and show the student's cumulative GPA using the grading rules from the SRMS database.
- **APR-4:** The system shall group academic records by semester so they are easy to read on a phone screen.
- **APR-5:** The system shall send a notification to the student's device within 5 seconds of a grade update.

3.5 Payment Management

3.5.1 Description and Priority

The Payment Management feature handles the tuition fees for students who are required to pay them. This feature remains hidden for Guyanese residents but is visible for Foreign and CARICOM nationals, therefore it is a conditional feature. For fee-paying students, course registration is considered "Provisional" until payment is verified.

The system will support two modes of payment:

- Automatic: Using MMG (Mobile Money Guyana) for instant clearance.
- Manual: Uploading proof of payment (e.g. bank transfer receipts) for manual staff verification.

Priority: Medium (Planned for Phase 2)

Per client feedback, this feature is critical for the system's long-term goals but may not be included in the initial launch. However, the requirements are specified here to ensure the architecture supports it.

3.5.2 Stimulus/Response Sequences

- **Stimulus:** A foreign student navigates to the Finances tab.
- **Response:** The system displays the current invoice and outstanding balance.
- **Stimulus (Option A):** The student selects "Pay with MMG."
- **Response:** The system initiates a secure transaction via the MMG API. Upon success, the system automatically updates the status to Cleared and restores access to grades.
- **Stimulus (Option B):** The student selects "Upload Receipt" (for bank transfers).
- **Response:** The system displays payment options (e.g., Pay Online or Upload Bank Receipt).
- **Response:** The system allows the user to take a photo or upload a file. The system flags the account for Manual Verification by Registry staff.
- **Response:** Once staff verifies the receipt manually in the backend, the app sends a notification: "Payment Verified."

3.5.3 Functional Requirements

- **PM-1:** The system shall hide all payment screens for users identified as Guyanese residents.
- **PM-2:** The system shall integrate with the MMG (Mobile Money Guyana) to allow automated payment processing.
- **PM-3:** The system shall automatically update the student's financial status to Cleared immediately after a successful MMG transaction.
- **PM-4:** The system shall provide a "Proof of Payment" upload feature (in JPEG, PNG, or PDF format) for students paying via bank transfer or other manual methods.
- **PM-5:** The system shall set the status of manual uploads to Pending Verification until a staff member approves the transaction in the backend.
- **PM-6:** The system shall block access to the Academic Records and Transcript features for foreign students with an Uncleared financial status.
- **PM-7:** The system shall keep a log of all transactions and uploaded receipts for auditing purposes.

3.6 Use Case Specification

Use Case ID: UC-01
Use Case Name: Register for Courses

Field	Description
Actors	Student, SRMS API
Description	This use case describes the process where a student selects, validates, and submits their course registration for the current semester.
Preconditions	1. The student is logged into the app. 2. The registration period is open 3. The student has no holds preventing registration.
Main Flow	1. The Student opens the Registration screen. 2. The System retrieves available courses, prerequisites and the student's current status. 3. The Student adds desired courses to their provisional list. 4. The System validates the selection against prerequisites and programme rules. 5. The Student navigates to the “Preview” screen. 6. The System displays the selected courses and total credits. 7. The Student confirms and taps “Submit.” 8. The System sends the data to the SRMS API. 9. The System displays a success message and the updated registration status (e.g. “Submitted”).
Postconditions	The student's registration is recorded in the central SRMS database, and the status is updated to “Submitted” pending approval.
Exceptions	**4a. Validation Failed:** The system detects a missing prerequisite. It displays an error message and prevents the course from being added. **8a. Submission Failed:** The API is unreachable. The system displays a “Connection Error” message and saves the draft locally.

Use Case ID: UC-02
Use Case Name: Apply to University

Field	**Description**
Actors	Prospective Student, SRMS API
Description	This use case allows a new user to complete and submit an application for a university programme, including document uploads.
Preconditions	The user has created an applicant account and is logged in.
Main Flow	1. The Student selects "Apply to UG." 2. The Student completes the Personal Information and Address forms. 3. The System validates and saves the data. 4. The Student enters Education and Employment history. 5. The Student selects a Programme from the available list. 6. The Student uploads required support documents (IDs, Certificates). 7. The Student reviews the full application summary. 8. The Student confirms submission. 9. The System transmits data to the backend and displays a confirmation message.
Postconditions	The application is created in the SRMS database with a status of Submitted or Under Review.
Exceptions	**6a. Upload Error:** The file format is unsupported or too large. The system displays an error and prompts for a valid file. **9a. Duplicate Application:** The system detects an existing application for this user. It displays a notification and redirects the user to the status tracking screen.

Use Case ID: UC-03
Use Case Name: View Timetable

Field	**Description**
Actors	Student, SRMS API
Description	This use case allows a registered student to view their class schedule for the current semester.
Preconditions	1. The student is logged in. 2. The student is officially enrolled in at least one course.
Main Flow	1. The Student selects the "Timetable" option. 2. The System retrieves the student's enrolled courses and associated time/venue data. 3. The System displays the schedule in a weekly format. 4. The Student toggles between weekly and daily views. 5. The Student taps a specific class for details.
Postconditions	The student has viewed their schedule. No data is changed.
Exceptions	**2a. No Classes:** If the student is not registered, the system displays "No registered courses found." **3a. Hybrid Class:** If a class is marked "Hybrid," the system displays a note advising the student to contact the lecturer.

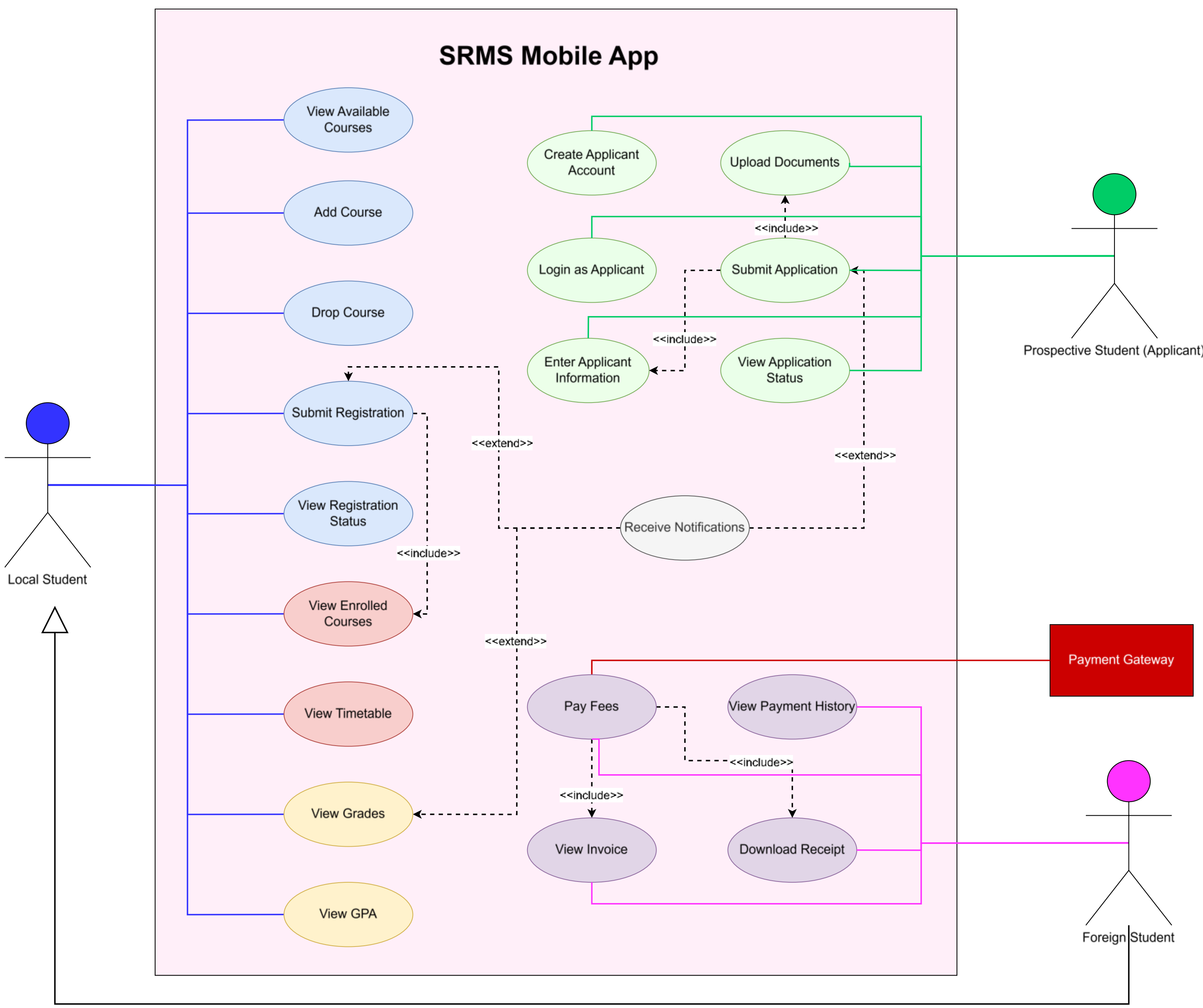
Use Case ID: UC-04
Use Case Name: View Academic Record

Field	**Description**
Actors	Student, SRMS API
Description	This use case allows a student to view their historical grades, current semester results, and cumulative Grade Point Average (GPA).
Preconditions	1. The student is logged into the application. 2. Grades have been published by the Registry.
Main Flow	1. The Student selects "My Grades" from the home dashboard. 2. The System requests the student's academic history from the SRMS database. 3. The System calculates and displays the current cumulative GPA. 4. The System lists all completed semesters. 5. The Student taps on a specific semester. 6. The System expands the view to show courses taken in that semester and the final letter grade received for each.
Postconditions	The student has successfully viewed their academic performance. No data is modified.
Exceptions	**2a. Financial Hold:** If the student has a financial hold that blocks grade viewing, the system displays a notification: "Grades are unavailable due to an outstanding balance. Please contact the Bursary."

Use Case ID: UC-05
Use Case Name: Make Tuition Payment

Field	**Description**
Actors	Student (Foreign/CARICOM), MMG System, Registry Staff
Description	This use case describes how a non-resident student pays tuition fees using MMG or manual receipt upload to regain access to grades.
Preconditions	1. The user is logged into the app. 2. The user is marked as Foreign or CARICOM 3. An invoice with an outstanding balance exists.
Main Flow	**A. MMG Payment (Automatic)** 1. Student selects "Pay with MMG." 2. System redirects to MMG interface. 3. Student completes transaction. 4. System receives success signal. 5. System updates status to Cleared and unlocks grades **B. Manual Payment (Upload)** 1. Student selects "Upload Receipt." 2. Student uploads image of bank receipt. 3. System saves image and sets status to Pending Verification. 4. Registry Staff (Backend) verifies receipt. 5. System sends "Payment Approved" notification.
Postconditions	The student's balance is updated. If paid via MMG, academic blocks are removed immediately.
Exceptions	**A1. MMG Failed:** Transaction fails. System displays error and retains blocks. **B1. Receipt Rejected:** Staff rejects the uploaded image (e.g., blurry). System sends "Resubmit Receipt" notification.

3.7 Use Case Diagram



4. Other Nonfunctional Requirements

4.1 Performance Requirements

- **NFR-1:** The SRMS Mobile Application shall be compatible with iOS, the latest 3 major versions, and Android API 26 and higher.
- **NFR-2:** The SRMS Mobile Application shall support simultaneous user requests without noticeable slowdown by ensuring that 90% of all SRMS information transactions have a Transaction Response Time (TRT) of less than 900 ms, and that 95% of backend API calls respond within 2 seconds under both normal and peak loads.
- **NFR-3:** The main pages, which are Dashboard, Course Registration, Academic Record, Timetable and Payment Status, shall fully load within 4 seconds when the device has a stable network connection with a minimum download speed of 5Mbps.
- **NFR-4:** All actions initiated by the user on their account in the SRMS mobile application shall be completed and reflected on both the SRMS mobile application and the existing SRMS within 5 seconds of the user confirming the action.
- **NFR-5:** The SRMS mobile application shall limit payload sizes for data retrieval requests to 2MB for slow networks.
- **NFR-6:** Push notifications from the SRMS mobile application shall be successfully delivered to users within 5 seconds of the corresponding event occurring on the existing SRMS.
- **NFR-7:** The system shall maintain NFR-3 on all core screens, supporting a minimum of 1,000 concurrent users during peak periods.

4.2 Safety Requirements

- **NFR-8:** The system shall use automated data backup or transaction logging to prevent loss during crashes or unintended modifications.
- **NFR-9:** The system shall verify that all transmitted data follows strict validation rules defined by the existing SRMS before any data write is executed, rejecting any invalid or incomplete submissions and correcting the user.
- **NFR-10:** The system shall use idempotency keys to track submissions of payments and registrations to the university's servers, which ensures that if the connection is interrupted, the transaction is processed exactly once, preventing duplicate submissions.
- **NFR-11:** The SRMS Mobile application shall display a confirmation prompt requiring the user to tap "Confirm" or "Submit" a second time before sending the irreversible request for permanent actions.
- **NFR-12:** The system shall store temporarily cached data securely and clear data upon user log-out or successful submission.

- **NFR-13:** The system shall ensure that sensitive data is not exposed in system logs or crash reports.

4.3 Security Requirements

- **NFR-14:** The system shall not store user passwords locally. User credentials shall be transmitted securely via HTTPS to the University's central authentication service to obtain a secure access token for session management.
- **NFR-15:** The system shall automatically terminate an active user session and require the user to log in again if no activity is detected for 30 minutes.
- **NFR-16:** All data communication between the SRMS mobile application and the university server shall use HTTPS with TLS 1.2 or higher version, and the system shall enforce SSL certificate pinning to verify the server connection and prevent man-in-the-middle attacks.
- **NFR-17:** The system shall not permanently save sensitive data, including grades and personal information, on the mobile device, and any temporary data that is required shall be saved only in an encrypted format.
- **NFR-18:** The system shall manage payments entirely by the University of Guyana's approved third-party secure payment gateway, ensuring the SRMS mobile application never stores sensitive financial details.
- **NFR-19:** The system shall be audited to verify compliance with Guyana's Cybercrime Act and all specific university data privacy policies before being released to students.
- **NFR-20:** The system shall enforce role-based access control so that students, lecturers, and administrators access only their authorized features.

4.4 Software Quality Attributes

- **NFR-21:** The SRMS mobile application shall allow a user to navigate any core feature from the main dashboard within a max of 5 taps.
- **NFR-22:** The system shall maintain 99% uptime, excluding scheduled maintenance.
- **NFR-23:** The system shall use structured exception handling, API response validation, and retry logic to detect, manage, and recover from unexpected errors without terminating the application.
- **NFR-24:** The system shall adapt to various screen sizes.
- **NFR-25:** The system shall utilize a modular architecture and adhere to documentation standards to facilitate efficient updates and future feature expansion.

4.5 Business Rules

- **NFR-26:** The system shall allow students to access only services associated with their academic role, such as viewing grades, registration, and fee statements.
- **NFR-27:** The system shall enforce institutional course registration rules, including prerequisites, credit limits, and deadlines.
- **NFR-28:** The system shall allow only authorized academic staff to upload or modify grades, attendance, or course information.
- **NFR-29:** The system shall log all transactions (profile updates, course applications, requests) for auditing and transparency.
- **NFR-30:** The system shall follow all institutional regulations and workflows defined for the existing SRMS system to ensure platform consistency.

Collaborative Summary

Division of Labour:

Name	Tasks	Signature
Kaylee Carter	Other Nonfunctional Requirements (Performance Requirements, Safety Requirements, Security Requirements)	K. Carter
Khaleel Khan	Other Nonfunctional Requirements (Security Requirements, Software Quality Attributes, Business Rules)	[illegible]
Ravindra Persaud	Introduction & System Features (Course Registration, Student Application, Timetable Viewing)	R. Persaud
Vimla Persaud	Overall Description & System Features (Use Case Diagram)	Vimla Persaud
Tulsidai Singh	System Features (Academic Progress and Records, Payment Management, Use Case Specifications)	T. Singh

Collaboration Strategies and General Experiences:

Our group used a shared Google Doc for real-time collaborative writing and a WhatsApp group for daily communication and coordination. This strategy worked well, as the shared document allowed everyone to see progress and contribute easily. Regular check-in meetings were held via Google Meet to ensure everyone was aligned.

Challenges and Resolutions:

Our main challenge was ensuring a consistent writing style and tone across the document, since each section was written by a different person. We handled this by designating one member as the final compiler, responsible for reviewing the entire document and making minor edits to ensure it read as a single, cohesive report.